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Analyzing the Effectiveness of Transfer Learning and Custom CNN Approaches in Corn Disease Classification

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Abstract—**Agriculture has a meaningful contribution to both food and livelihood. Corn belongs to the major food crop in Bangladesh. However, its productivity is greatly hampered by leaf diseases—Common Rust, Gray Leaf Spot, and Northern Leaf Blight. The traditional manual diagnosis process is slow, laborious, and inefficient for farmers who do not have the necessary resources. To address these challenges, this paper proposes an automated, deep learning based system for accurate and timely classification of corn leaf diseases, empowering farmers with scalable and affordable solutions. The PlantVillage dataset, comprising images of diseased and healthy corn leaves, is utilized for model development. Four state-of-the-art pretrained convolutional neural networks-InceptionV3, ResNet152V2, DenseNet201, and EfficientNetB4 are fine-tuned via transfer learning, achieving accuracies of 99.49%, 99.34%, 99.12%, and 98.76%, respectively. Additionally, a lightweight CNN is designed, with only three convolutional layers and no frozen layers. This model reached an accuracy of 95.76%. Although its accuracy is slightly lower, its compact design, faster performance, and lack of reliance on external weights make it perfect for offline use on mobile devices or edge devices in resource-limited rural areas. The results show that using transfer learning with deep CNNs achieves top accuracy, while the custom network balances performance, adaptability, low hardware demands, and interpretability. The proposed framework allows for early disease classification. This approach reduces yield loss and helps improve long-term food security in Bangladesh.**

Keywords— PlantVillage, deep learning, transfer learning, find-tuned, CNN, InceptionV3, ResNet152V2, DenseNet201, EfficientNetB4, lightweight, custom .

I. INTRODUCTION

Corn is a grain that connects farming ,food,and industry.It has spread to districts like Dinajpur, Bogura, and Lalmonirhat to satisfy the rising demand in Bangladesh. However,diseases like Common Rust,Gray leaf Spot,and Northern leaf Blight often threaten its productivity.These diseases can cause significant yield losses,hurting farmers incomes and the countries food security.In rural Bangladesh, farmers and agricultural officers mainly rely on manual inspection for disease diagnosis [8]. Therefore, there is a strong need for an intelligent, automated,and accessible solution for early and accurate disease classification. Recent advancement in computer vision and deep learning have made automated crop disease diagnosis more practical. Convolutional Neural Networks or CNNs [4], especially when combined with transfer learning,have shown great potential in image based classification tasks, including plant disease detection. By using pretrained models that have trained on large datasets (ImageNet), transfer learning allows for accurate disease classification even with limited agricultural data.This research aims to develop deep learning based system for classifying corn leaf diseases using the corn subset of the publicly available PlantVillage dataset [1].This dataset has thousands of images of both healthy and diseased corn leaves. This study employs four top performing pretrained CNN architectures such as EfficientNetB4, DenseNet201, InceptionV3, ResNet152V2 [1]. This models are fine-tuned to identify corn-specific disease patterns by modifying their classification layers and optimizing them for this task. Alongside evaluating these pretrained models, a custom CNN architecture has been created to provide a better performance while keeping computational needs low. The main goals of this study are to apply and evaluate several pretrained CNN models for classifying corn leaf diseases. Design and test a lightweight custom CNN model for practical use .This research covers classifying corn leaf diseases from images, optimizing deep learning structures for this task, and assessing their performance in speed and accuracy. The models built here provide a strong foundation for future integration into user-friendly tools for farmers. By automating the disease classification process, this research aims to give farmers timely information, lower crop losses, reduce pesticide misuse, and encourage sustainable farming. The findings add to the growing field of AI in agriculture and show how Deep learning [2] can significantly impact on food security in Bangladesh.

II. RELATED WORK

Recent advances in deep learning and computer vision have significantly enhanced the automatic detection and classification of corn plant diseases. One study proposed an end-to-end deep learning framework combining.
EfficientNetB0 and DenseNet121 through feature fusion, achieving an accuracy of 98.56% on the PlantVillage corn leaf dataset. This approach demonstrated improved classification performance over individual pretrained models [1].

A hybrid deep learning model combining CNNs,ResNet, and EfficientNetB3 was proposed for real-time detection of rice and potato leaf diseases. Using a Decision Tree classifier on features extracted from a comprehensive dataset, the model achieved 98.5% accuracy in classifying key diseases. This approach offers precise and interpretable disease detection for sustainable agriculture [2].

This study addresses the challenge of limited processed data for plant disease diagnosis by developing comprehensive datasets for rice, wheat, and maize, focusing on common bacterial and fungal diseases. Eight fine-tuned deep learning models were evaluated, with Xception and MobileNet variants achieving the highest accuracies across different crops. Additionally, a novel CNN model trained from scratch demonstrated strong performance, achieving testing accuracies above 97% on all three datasets. This work highlights the effectiveness of customised datasets and models for accurate food grain disease detection [3].

Additionally, a computer vision pipeline incorporating classical image segmentation followed by a lightweight CNN classifier was used for corn leaf disease detection on field-collected images, achieving 93% accuracy on proposed model. This work highlighted the effectiveness of combining traditional image processing techniques with deep learning models for practical deployment in resource-limited agricultural environments [4].

This research developed a corn leaf disease detection model using a Kaggle dataset. Preprocessing included resizing and HSV colour segmentation,with feature extraction via Hu Moments, Haralick,and Histogram methods. Support Vector Machine(SVM) achieved the best accuracy of 98.53%, demonstrating reliable disease identification to improve corn health and productivity [5].

This study presents a hybrid model combining CNN based feature extraction with an XGBoost classifier for corn leaf disease detection. Three model is deployed via cloud-based APIs supporting both web and mobile platforms, enabling accessible real-time diagnosis. Performance comparisons with other deep learning models demonstrate the proposed system's competitive accuracy and efficiency for practical agricultural applications [6].

A CNN-based method was proposed for corn disease identification using wavelet and histogram thresholding for image preprocessing and segmentation. A five-layer deep CNN model was trained with transfer learning and optimized via gradient and mean square error methods. The system achieved a 96.8% accuracy in identifying common corn diseases, demonstrating practical applicability for corn management [7].

Together, these studies show that using different feature extraction techniques and transfer learning methods can greatly improve the accuracy and efficiency of corn leaf disease classification systems. However, more research is needed to make these models suitable for real-time use in the field, particularly in resource-limited areas.

III. METHODOLOGY

It includes several key stages A, B, C and D . Keras and Tensorflow are used as backend libraries to develop the deep learning models and all pictures used in this study except dataset. Fig.1 shows methodology of this study.

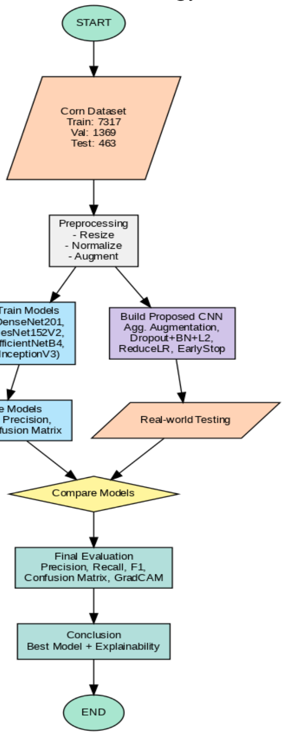

Fig.1. Methodology flow diagram.

A. Dataset Description

The dataset has clear backgrounds and moderate noisy, & the images are in RGB format but were not properly labeled [6]. To prepare the dataset, an 80:15:5 ratio was used to split it into training, validation and testing sets.For clarity, images were labeled as (Corn_GrayLeaf Spot_4.jpg) in Fig.2. Class imbalance was reduced by selecting an equal number of images for each class to ensure balanced representation. To maintain consistency, the same dataset structure and class distribution were used across all models both pretrained and custom models for fair and unbiased comparison. Table I shows whole data collection summary.

Fig.2. Example images from the dataset (kaggle) with respective class labels.

TABLE I. DATA COLLECTION SUMMARY FOR THIS STUDY.

Category	Details
Source dataset	PlantVillage(approx. 54,000 images total).
Corn images used	9149 real images(4 classes: Common Rust, Gray Leaf Spot, Northern Leaf Blight, Healthy).
Train,validation,test set	7317, 1369 , 463 images.
Train set after augmentation	36,585 images.
Image conditions	Clear background,low noisy,properly labeled,augmented,preprocessed.

Although the dataset was balanced, differences between lab images and field images may affect how well the model performs on new data. In real-world situations, lighting, background, leaf position, and camera quality can vary significantly, which may lead to wrong predictions. Using more diverse and real-world images in the future will help make the model robust and more generalizable.

B. Preprocessing and Augmentation

To prepare, clean, and standardize the dataset for training, several preprocessing steps were applied. All images were resized according to the input requirements of the respective CNN models :

- InceptionV3 :299 × 299
- ResNet152V2 and DensNet201: 224 × 224
- EfficientNetB4: 300 × 300
- Hybrid(InceptionV3-ViT): 224 × 224

Each pretrained model has an optimal input size selected to balance feature extraction and computational efficiency. Using the appropriate input size ensures effective pattern capture without unnecessary resizing, thereby maintaining accuracy. Next, all pixel values were normalized to the [0,1] range by rescaling using a factor of 1/255. To enhance the diversity and volume of training data and improve generalization, data augmentation techniques were applied to the training set, expanding it from 7,317 to 36,585 images. The augmentation strategies included rotation (40 degrees), width & height shift, shearing(0.2),zooming(0.7 to 1.3), brightness adjustment(0.6 to 1.4), channel shifting (up to 20.0 units), fill mode (nearest), and random horizontal flipping. These transformations help simulate real-world variations in lighting, orientation, and leaf positioning, improving model robustness and reducing the risk of overfitting across all models.

C. Conceptual background

As the foundation of this study, CNNs in Fig. 3 are highly effective at automatically extracting meaningful features from leaf images, which is essential for identifying subtle differences between healthy and diseased plant tissues. Fig. 4 shows a Sample pictures of feature extraction by DenseNet201.Convolutional layers use small filters to detect local patterns like edges, textures, and shapes, creating feature maps that capture increasingly complex patterns as the network goes deeper. Convolutional operation follows the equation (1). Pooling layers reduce spatial dimensions to lower computational load while retaining key details, and fully connected layers combine these features to make the final predictions.By using pretrained CNNs on corn leaf images and visualizing the convolutional outputs shows how early layers capture simple features whereas the deeper layers detect more complex patterns, improving accuracy and helping explain how the model makes its decisions.

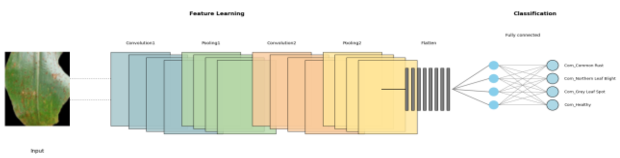

Fig. 3. Basic Convolutional Neural Network (CNN) Architecture.

$$(I \times K)(x,y) = \sum I(x+i, y+i).K(i,j) \quad (1)$$

- I =input image patch
- K=kernel(filter)
- the sum is over the size of the kernel

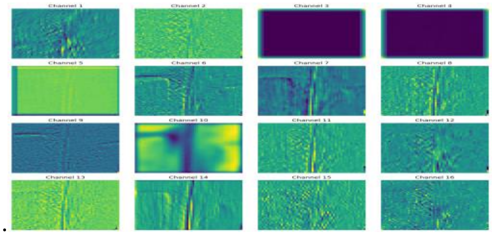

Fig. 4. Sample pictures of feature extraction by DenseNet201.

D. Transfer Learning approach for pretrained models, hybrid model and proposed model

The four pretrained base models [1] highlighted in this study have unique architectural strengths for image classification. To adapt these models for corn leaf disease classification, their original top layers are removed and replaced with custom dense layers designed for four-class output. Here, the pretrained model architectures work as feature extractors while the added custom dense layers perform the final classification of corn leaf diseases. Additionally, selected deeper layers were fine-tuned, and hyperparameters were optimized to enhance accuracy with 20 epochs. These modifications in Table II helped retain the benefits of pretrained feature extraction while enabling the models to learn disease specific patterns effectively. In all models, a GlobalAveragePooling2D layer was applied to reduce feature dimensions, followed by dense layers consisting of 512 and 256 units with ReLU activation. Dropout(0.4-0.5) for regularization and BatchNormalization to stabilize training, The Adam optimizer and Categorical Crossentropy loss were used with a batch size of 32.To prevent overfitting, EarlyStopping and ReduceLROnPlateau were applied. Fine-tuning in all models was performed on the last layers as per keras's layer-wise structure.

TABLE II. SPECIFIC MODIFICATIONS FOR MODELS

Model	Specific Modifications
InceptionV3	Fine-tuning was applied to the last 100 trainable layers, was trained using Adam (LR=1e-4), with EarlyStopping and learning rate reduction to ensure stable and efficient convergence.
ResNet152V2	Only the last 50 layers were fine-tuned. The learning rate was set to 1e-4, and Early-Stopping was triggered after 6 epochs of no improvement, while LR was reduced after 3 epochs.
DensNet201	Fine-tuning on the last 100 layers;L2regularization added to dense layers for improved generalization. The learning rate was 2e-4.
EfficinetNetB4	Fine-tuned last 100 layers, Learning rate started at 1e-4 and reduced by factor of 0.5 on plateau.
InceptionV3-ViT(Hybrid)	Adam (lr=0.0001) was used with categorical crossentropy, combining InceptionV3(last 100 layers trainable) and ViT (patch=16, heads=4, dim=64), with Dropout (0.5,0.4) and callbacks for stable training..

Proposed CNN model Architecture:
Conv2D (32 filters, 3×3) + ReLU ⟶ BatchNormalization ⟶ MaxPooling(2×2)⟶ Dropout(0.25)
Conv2D (64 filters, 3×3) + ReLU ⟶ BatchNormalization ⟶ MaxPooling(2×2)⟶ Dropout(0.25)
Conv2D (128 filters, 3×3) + ReLU ⟶ BatchNormalization ⟶ MaxPooling(2×2)⟶ Dropout(0.25)
Flatten
Dense(256 units) + ReLU⟶ Dropout(0.5)
Dense(4 units) + Softmax

- Optimizer: Adam (learning rate = 0.0001)
- Loss: Categorical Crossentropy
- Metrics: Accuracy
  (Input size : 224×224×3)

TABLE III. Model Summary

Total parameters	Trainable parameters	Non-trainable parameters
22,246,596(84.86 MB)	22,246,148(84.86 MB)	448(1.75 KB)

Convolutional layers, BatchNormalization, Maxpooling, and Dropout layers work together to extract important features from the images, reduce unnecessary information, and protect the model from overfitting. The Flatten and Dense layers analyze these features and help classify the image into one of the four categories, while the final softmax layer gives the probability of the image belonging to each class so that the model can make its decision. The same preprocessing and augmentation techniques are used in the pretrained models. The model architecture is simple and lightweight, consisting of only three convolutional layers followed by batch normalization, dropout and two dense layers. This study uses transfer learning-based CNNs for corn disease classification [1]. Here, Table IV shows proposed model data pipeline analysis.

TABLE IV. PROPOSED CNN MODEL & DATA PIPELINE ANALYSIS

Component	Feature added	Purpose/Impact
Conv2D Layers (32, 64,128 filters,3x3)	Extracts hierarchical features .	Learns spatial patterns at increasing complexity.
BatchNormalization	Applied after each convolutional layer.	Stabilizes & accelerates training, reduces internal covariate shift.
MaxPooling2D	Reduces spatial dimensions.	Controls overfitting, lowers computational cost.
Dropout(0.25/0.5)	Randomly drops neurons during training .	Prevents overfitting, improves generalization.
Dense(256) + ReLU	Fully connected layer to combine features.	Learn complex relationships before classification.
Dense(4) + Softmax	4- class output	Provides class probabilities for final prediction.
Adam Optimizer (lr=0.0001)	Carefully tuned small learning rate.	Ensures stable convergence, avoids overshooting.
Categorical Crossentropy loss	Suitable for multi-class problem.	Measures error between prediction and true label.
Batch Size = 32	Mini-batch size during training.	Balanced between memory efficiency and stable gradient updates.

In total 22.2 million parameters are there shown in Table III , all of which are trainable, providing full flexibility for fine-tuning on custom or local datasets without relying on external pretrained weights. In contrast, many pretrained models freeze a large portion of parameters, limiting adaptability. Moreover, the absence of dependency on large-scale datasets like ImageNet reduces complexity, data mismatch issues, and storage requirements. The simplicity of the model architecture also makes it more interpretable. Despite of achieving marginally lower accuracy, the proposed model strikes a balance between performance, scalability, and deployment readiness making it a practical solution for real time, on-field corn leaf disease detection in low-resource settings.

IV. RESULT AND COMPARATIVE ANALYSIS

When comparing models whether between pretrained architectures or with the proposed model, accuracy alone isn't enough. While it shows the overall correctness, it doesn't reflect how well the model performs for each specific disease class. To get a clearer picture, metrics like precision (2), recall (3), F1-score (4), and ROC-AUC were also used, as they offer deeper insights into the reliability and class-wise performance of the models.
Precision: Measures the positive predictions made by the model are actually correct.

$$\textbf{Precision} = \frac{\text{T.P.}}{\text{T.P.+ F.P.}} \tag{2}$$

Recall: Measures the actual positive cases the model correctly identified.

$$\textbf{Recall} = \frac{\text{T.P.}}{\text{T.P.+ F.N.}} \tag{3}$$

Confusion Matrix**:** Summarizes correct and incorrect predictions by comparing actual vs predicted classes.

F1-Score: F1 score is the harmonic mean of precision and recall.It balances both precision and recall in a single metric.

$$\textbf{F1} - \textbf{score} = \frac{\textbf{2} \times \textbf{precision} \times \textbf{recall}}{\textbf{precision} + \textbf{recall}} \tag{4}$$

The ROC curve visually represents how a trained model performs across different threshold values by plotting the True Positive Rate(TPR) against the False Positive rate (FPR) .It helps identify the threshold where the TPR is high and the FPR is low–ideally, close to 1 on the TPR axis. This optimal point reflects the threshold at which the model performs best in distinguishing between classes. This section presents the results of training both the pretrained models and the proposed CNN model for corn leaf disease classification. Initially trained with basic settings for 10 epochs, the models showed moderate accuracy. To improve performance, fine-tuning was applied using techniques like layer-unfreezing, Optimizer, L2-regularization, Dropout, BatchNormalization, and adaptive learning rates. After training for 20 epochs with these adjustments, all models showed significant improvement. For this study, Sample results obtained using the pretrained InceptionV3 model shown in Fig.5 & Fig.6 also proposed model shown in Fig.

7, Fig. 8, Fig.9 along with the corresponding accuracy & loss graph, confusion matrix and ROC curve(20 epochs).

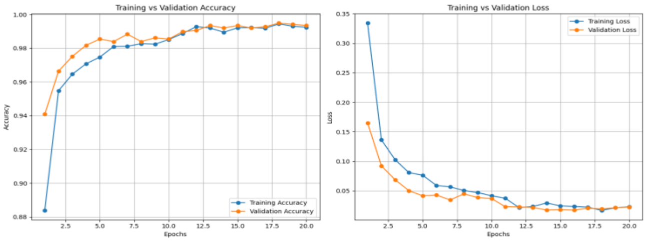

Fig.5. InceptionV3(Accuracy & Loss Graph)

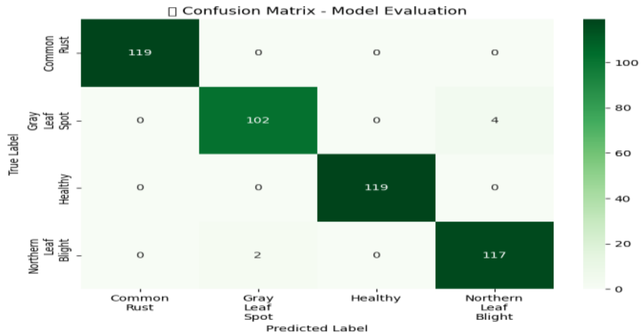

Fig. 6. InceptionV3 Confusion matrix (20 epochs)

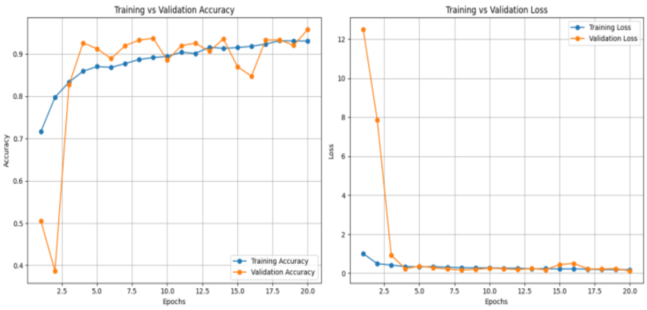

Fig. 7. Proposed Model (Accuracy & Loss Graph)

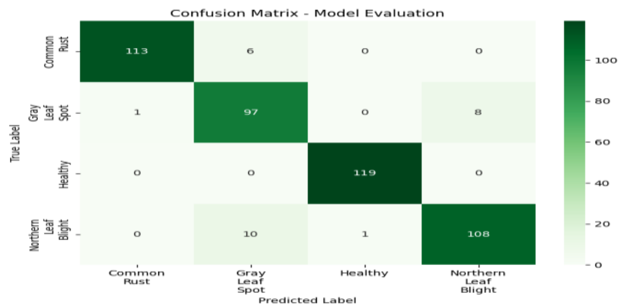

Fig. 8. Proposed Model (Confusion matrix)

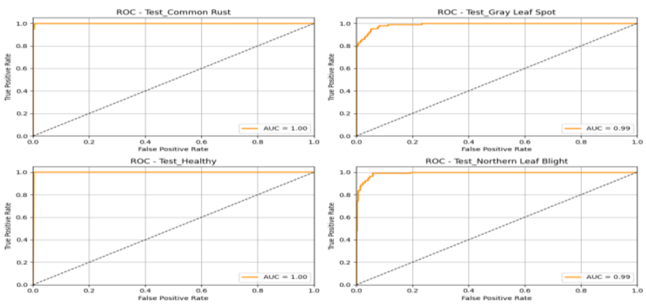

Fig. 9. Proposed Model (ROC curve)

Models result are shown in Table V, Table VI, Table VII:

TABLE V. INCEPTIONV3 PERFORMANCE SUMMARY

Crop	Precision	Recall	F1-Score	Support
Common Rust	1.00	1.00	1.00	119
GrayLeaf spot	0.98	0.96	0.97	106
Healthy	1.00	1.00	1.00	119
NorthernLeaf Blight	0.97	0.98	0.97	119

TABLE VI. PROPOSED MODEL PERFORMANCE SUMMARY

Crop	Precision	Recall	F1-Score	Support
Common Rust	0.99	0.95	0.97	119
Gray Leaf spot	0.86	0.92	0.89	106
Healthy	0.99	1.00	1.00	119
Northern Leaf Blight	0.93	0.91	0.92	119

TABLE VII. Hybrid MODEL PERFORMANCE SUMMARY

Crop	Precision	Recall	F1-Score	Support
Common Rust	0.99	1.00	1.00	119
Gray Leaf spot	0.96	0.96	0.96	106
Healthy	1.00	1.00	1.00	119
Northern Leaf Blight	0.97	0.96	0.96	119

After training, the models achieved validation accuracies of 99.49%(InceptionV3), 99.34%(ResNet152v2), 99.12% (DenseNet201), 98.76%(EfficientNetB4), respectively. The Proposed model achieve 95.76% validation accuracy using augmentation, regularization, balanced architecture, tuned hyper parameters and testing Accuracy 94.38% while keeping overfitting in check.

Key Comparisons :

In Conclusion,InceptionV3 and ResNet152V2 are best where computational resources are available on the other hand, proposed CNN offers a cost-effective alternative with moderate accuracy, showing in Table VIII which is feasible for real time use in the field.While pretrained models achieve higher accuracy than the proposed CNN but the custom model's simplicity and scalability make it a suitable choice for resource-limited agricultural settings in Bangladesh. Although this study employs a hybrid model (InceptionV3-ViT) which achieves less accuracy than some observed pretrained models because it typically requires even larger datasets, longer training, more GPU or TPU resources, and are harder to interpret due to their complex black-box architectures.

TABLE VIII. COMPARATIVE ANALYSIS WITH PRETRAINED AND PROPOSED MODEL

Model	Val. Accuracy	Val. Loss	Test Accuracy	Training Time (Hours)
InceptionV3	99.49%	0.0193	98.70%	3.57 hr
ResNet152V2	99.34%	0.0186	98.70%	3.75 hr
DenseNet201	99.12%	0.1275	98.92%	2.33 hr
EfficientNetB4	98.76%	0.0256	98.49%	2.17 hr
Proposed CNN	95.76%	0.1221	94.38%	2.50 hr
Hybrid(Inception V3-ViT)	98.69%	0.0328	98.06%	4.53 hr

The best model was evaluated on the corn dataset, and it successfully generated accurate predictions.A sample output in Fig.10 is illustrated below to demonstrate the model's performance.

Fig. 10. Predicted output of the proposed model on test data.

To enhance model interpretability, Grad-CAM was applied to visualize the regions of corn leaves in Fig. 11 that contributed most to disease classification. The resulting heatmaps are presented below .

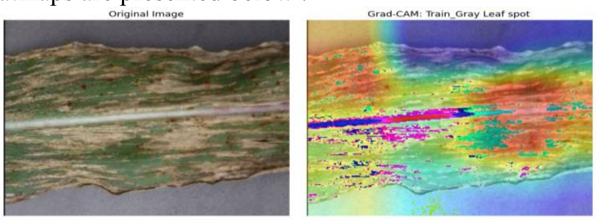

Fig 11. Grad-CAM visualizations showing activated regions for correctly predicted corn leaf disease samples.

TABLE IX. WORKING ENVIRONMENT & SPECIFICATION

Working Environment	**Specification**
Development Environment	Kaggle kernel
GPU	NVIDIA Tesla T4

V.FUTURE WORK AND CONCLUSION

Several directions can be explored to further enhance for this research:

Combining both approaches, such as using InceptionV3 to the custom CNN , to narrow the performance gap. Optimization of the model for web Application and mobile app in agricultural fields. Development of a multi-crop disease classification model (Rice,Corn,Potato) to handle different crops in a single system. In this study, deep neural network architectures, e.g., InceptionV3, ResNet152V2, DenseNet201, EfficientNetB4 using transfer learning, were used to classify the corn leaf disease with incredible results.The developed light CNN model also achieved satisfactory results which demonstrated it could be real time applied in resource-limited agricultural fields.

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